

Iniciemos el lab ENCRYPTED DATA LEAK

```
kali@kali: ~/Downloads/Labs/Facil/ENCRYPTED DATA LEAK
Session Actions Edit View Help
Aprende, simula, escala. Sin burocracia.

[*] Cargando imagen desde encrypted_data_leak.tar ...
[+] Imagen cargada exitosamente

[*] Iniciando laboratorio ...

[✓] Laboratorio desplegado correctamente.

LAB: ENCRYPTED DATA LEAK
DIFICULTAD: ★★★★★★ Facil (1 puntos)

IP del laboratorio: 172.17.0.3

CREDENCIALES SSH:
Usuario: estudiante
Password: student2026

COMANDO DE CONEXIÓN:
ssh estudiante@localhost -p 2223
O desde otra máquina:
ssh estudiante@<IP_DEL_HOST> -p 2223

Ingresa la flag (formato FLAG{...}):
```

Vemos que ya nos dan las credenciales de acceso y el comando de conexión

```
estudiante@encrypted_data_leak: ~
Session Actions Edit View Help
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[localhost]:2223' (ED25519) to the list of known hosts.
estudiante@localhost's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.19.14+kali-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ENCRIPTED DATA LEAK - LABORATORIO FORENSE

cd /investigacion
cat README.txt

estudiante@encrypted_data_leak:~$
```

Miremos si hay otros archivos antes de seguir las instrucciones dadas y ver qué tipo de acceso tenemos

```

estudiante@encrypted_data_leak:~$ id
uid=1000(estudiante) gid=1000(estudiante) groups=1000(estudiante)
estudiante@encrypted_data_leak:~$ whoami
estudiante
estudiante@encrypted_data_leak:~$ pwd
/home/estudiante
estudiante@encrypted_data_leak:~$ ls -al
total 28
drwxr-x--- 1 estudiante estudiante 4096 Jun 19 22:23 .
drwxr-xr-x 1 root root 4096 Jan 20 14:51 ..
-rw-r--r-- 1 estudiante estudiante 220 Jan 6 2022 .bash_logout
-rw-r--r-- 1 estudiante estudiante 3771 Jan 6 2022 .bashrc
drwx----- 2 estudiante estudiante 4096 Jun 19 22:23 .cache
-rw-r--r-- 1 estudiante estudiante 807 Jan 6 2022 .profile

```

Procedemos con las instrucciones del archivo README.txt

PARTE 1: GENERAR HASHES DE ARCHIVOS ACTUALES

Usando los comandos de la tarea, genera hashes de los 4 archivos:

MÉTODO 1 - MD5:

md5sum database_config.txt

md5sum api_credentials.txt

md5sum server_setup.txt

md5sum access_logs.txt

```

estudiante@encrypted_data_leak:/$ cd /investigacion/
estudiante@encrypted_data_leak:/investigacion$ md5sum database_config.txt
d6bb94b94ed8540a4d8d0b782dbd807a database_config.txt
estudiante@encrypted_data_leak:/investigacion$ md5sum api_credentials.txt
ade0d90fe6cc76d4c4b21e5e79201633 api_credentials.txt
estudiante@encrypted_data_leak:/investigacion$ md5sum server_setup.txt
e24a038385bf095bf97fadb47299c3fc server_setup.txt
estudiante@encrypted_data_leak:/investigacion$ md5sum access_logs.txt
16916c766b51bdc63d4f76a769bc93e2 access_logs.txt
estudiante@encrypted_data_leak:/investigacion$ █

```

MÉTODO 2 - SHA256:

sha256sum database_config.txt

sha256sum api_credentials.txt

sha256sum server_setup.txt

sha256sum access_logs.txt

```
estudiante@encrypted_data_leak:/investigacion$ sha256sum database_config.txt
e3bd97f28e61e8fb9d5a6c0dd62814b43201976aecb29d1581c1b9d718cabcf0 database_config.txt
estudiante@encrypted_data_leak:/investigacion$ sha256sum api_credentials.txt
089bf22d7c390efd8f72d3ccaa36d3641235b7d97aec23bb0688f33dc849424a api_credentials.txt
estudiante@encrypted_data_leak:/investigacion$ sha256sum server_setup.txt
355e17035bb8322cbf8c5cdfa30378c8db13359b8d1d56bf7ca097f111aa7210 server_setup.txt
estudiante@encrypted_data_leak:/investigacion$ sha256sum access_logs.txt
2f5c53e05e7dbd0bac06b01eddda89e49a747da298c5008d4be45180e0a61476 access_logs.txt
estudiante@encrypted_data_leak:/investigacion$
```

MÉTODO 3 - SHA512:

sha512sum database_config.txt

sha512sum api_credentials.txt

sha512sum server_setup.txt

sha512sum access_logs.txt

```
estudiante@encrypted_data_leak:/investigacion$ sha512sum database_config.txt
935a43fbeb74c8d67a4aa16c24ecd3024f783432c5b1c9908aced3d8d053cd816cedaac4075762321fa4817b8bd21d104851a67955c009d01e0a574c302f7b51 database_config.txt
estudiante@encrypted_data_leak:/investigacion$ sha512sum api_credentials.txt
dce00c6ddfc793b72824b0e50ad6b38435d082cb789611f8942ed7b48a801d8084e5a87b1a32c0f53ebdae55fdb85e79a108bf58f8739b2abea3c6b1fe620bea api_credentials.txt
estudiante@encrypted_data_leak:/investigacion$ sha512sum server_setup.txt
a14f09c8768e4449173fa3c2d189b9697d5854d49d730ab9b8a9328e898b8f6a1ad76e7d08f53f663828a09bf701f5a2b5e5ee30d3b07e0bb33218e2b293e5c2 server_setup.txt
estudiante@encrypted_data_leak:/investigacion$ sha512sum access_logs.txt
409c49afce76088ca9529f669d9ced12036e2fbd2009f41378bb887311005af4fde71328d28fc3ec1e1cd6cdc8c86ad956ebee6a08b47809381a8dff1944aca0 access_logs.txt
estudiante@encrypted_data_leak:/investigacion$
```

OTROS ALGORITMOS (opcional):

sha1sum archivo.txt

b2sum archivo.txt

PARTE 2: COMPARAR HASHES CON BASELINE

Usa el comando diff para comparar cada archivo:

COMPARACIÓN CON MD5:

diff <(cat baseline_hashes.txt | grep database_config) <(md5sum database_config.txt)

diff <(cat baseline_hashes.txt | grep api_credentials) <(md5sum api_credentials.txt)

diff <(cat baseline_hashes.txt | grep server_setup) <(md5sum server_setup.txt)

diff <(cat baseline_hashes.txt | grep access_logs) <(md5sum access_logs.txt)

```

estudiante@encrypted_data_leak:/investigacion$ diff <(cat baseline_hashes.txt | grep database_config) <(md5sum database_conf
ig.txt)
1c1
< 787236e0762c7727183c69a93e342341 database_config.txt
---
> d6bb94b94ed8540a4d8d0b782dbd807a database_config.txt
estudiante@encrypted_data_leak:/investigacion$ diff <(cat baseline_hashes.txt | grep api_credentials) <(md5sum api_credentia
ls.txt)
1c1
< c42718f784de529b113bb62fad269e3e api_credentials.txt
---
> ade0d90fe6cc76d4c4b21e5e79201633 api_credentials.txt
estudiante@encrypted_data_leak:/investigacion$ diff <(cat baseline_hashes.txt | grep server_setup) <(md5sum server_setup.txt
)
1c1
< faebc0aebd293701fddfb9d32ed6e60d server_setup.txt
---
> e24a038385bf095bf97fadb47299c3fc server_setup.txt
estudiante@encrypted_data_leak:/investigacion$ diff <(cat baseline_hashes.txt | grep access_logs) <(md5sum access_logs.txt)
estudiante@encrypted_data_leak:/investigacion$ █

```

COMPARACIÓN CON SHA256:

diff <(cat baseline_sha256.txt | grep database_config) <(sha256sum database_config.txt)

diff <(cat baseline_sha256.txt | grep api_credentials) <(sha256sum api_credentials.txt)

diff <(cat baseline_sha256.txt | grep server_setup) <(sha256sum server_setup.txt)

diff <(cat baseline_sha256.txt | grep access_logs) <(sha256sum access_logs.txt)

```

estudiante@encrypted_data_leak:/investigacion$ diff <(cat baseline_sha256.txt | grep database_config) <(sha256sum database_c
onfig.txt)
1c1
< f6249732c6b365d3be048f8ee7276bfe57c05456d7f6a43364aaff814d016fd9 database_config.txt
---
> e3bd97f28e61e8fb9d5a6c0dd62814b43201976aecb29d1581c1b9d718cabcf0 database_config.txt
estudiante@encrypted_data_leak:/investigacion$ diff <(cat baseline_sha256.txt | grep api_credentials) <(sha256sum api_creden
tials.txt)
1c1
< d3b2e1121b244f9d70d30de0c336c5812ccd3efa13722d73397071e6337ad456 api_credentials.txt
---
> 089bf22d7c390efd8f72d3ccaa36d3641235b7d97aec23bb0688f33dc849424a api_credentials.txt
estudiante@encrypted_data_leak:/investigacion$ diff <(cat baseline_sha256.txt | grep server_setup) <(sha256sum server_setup.
txt)
1c1
< 4d2ab85330fc2b9feab1b94f35f1cfd7e56a33c1a7dadf912b088fffb452e74 server_setup.txt
---
> 355e17035bb8322cbf8c5cdfa30378c8db13359b8d1d56bf7ca097f111aa7210 server_setup.txt
estudiante@encrypted_data_leak:/investigacion$ diff <(cat baseline_sha256.txt | grep access_logs) <(sha256sum access_logs.t
xt)
estudiante@encrypted_data_leak:/investigacion$ █

```

INTERPRETACIÓN:

- Si NO hay output = Hashes coinciden = Archivo NO modificado
- Si HAY output (diferencias) = Hashes diferentes = Archivo MODIFICADO

⚠ IMPORTANTE: Los 4 archivos fueron tocados, pero solo UNO contiene información crítica. Analiza bien los cambios.

PARTE 3: INSPECCIONAR ARCHIVOS SOSPECHOSOS

Una vez identifique qué archivo(s) fueron modificados,
LÉELOS para buscar información oculta:

cat database_config.txt

```
estudiante@encrypted_data_leak:/investigacion$ cat database_config.txt
# Database Configuration File
# Last modified: 2025-12-15 14:30:22
# Author: System Admin

DB_HOST=192.168.1.100
DB_PORT=5432
DB_NAME=production_db
DB_USER=admin
DB_PASSWORD=SecurePass2025!
MAX_CONNECTIONS=100
TIMEOUT=30
BACKUP_ENABLED=true
```

cat api_credentials.txt

```
estudiante@encrypted_data_leak:/investigacion$ cat api_credentials.txt
# API Credentials Repository
# Last updated: 2025-12-20 09:15:33
# Maintainer: DevOps Team

SERVICE_NAME=CloudStorage
API_KEY=sk_live_4f8a9b2c1d3e5f6g7h8i9j0k
API_SECRET=cs_secret_9z8y7x6w5v4u3t2s1r
REGION=us-east-1
ENDPOINT=https://api.cloud-storage.com/v2
RATE_LIMIT=1000
```

cat server_setup.txt

```
estudiante@encrypted_data_leak:/investigacion$ cat server_setup.txt
# Server Configuration
# Date: 2025-12-18 16:45:10
# Environment: Production

SERVER_NAME=web-prod-01
IP_ADDRESS=10.0.2.50
OS_VERSION=Ubuntu 22.04 LTS
SSH_PORT=22
HTTPS_PORT=443
FIREWALL=enabled
MONITORING=active
LAST_BACKUP=2025-12-19
```

cat access_logs.txt

```
estudiante@encrypted_data_leak:/investigacion$ cat access_logs.txt
# System Access Logs
# Generated: 2025-12-21 11:22:45
# Security Level: High

[2025-12-21 08:15:23] User: admin - Login: SUCCESS
[2025-12-21 09:30:12] User: carlos - Login: SUCCESS
[2025-12-21 10:45:55] User: maria - Login: FAILED
[2025-12-21 11:20:33] User: system - Action: BACKUP_COMPLETED
[2025-12-21 11:22:40] User: unknown - Login: FAILED
[2025-12-21 11:22:41] ALERT: Suspicious activity detected
[2025-12-21 11:22:42] DEBUG_MODE: FLAG{XXXXXXXXXXXXXXXXXXXX}
[2025-12-21 11:22:43] System restored to normal operation
```

Antes de introducir la bandera vamos a tratar de ubicarla y leer su contenido

```
estudiante@encrypted_data_leak:/investigacion$ find / -name "flag.txt" 2>/dev/null
/var/www/flag.txt
estudiante@encrypted_data_leak:/investigacion$ cat /var/www/flag.txt
FLAG{XXXXXXXXXXXXXXXXXXXX}
```

El contenido es el mismo, y el archivo que no se modifiko es el que contiene la bandera

Al introducir la bandera nos da resultado positivo

```
kali@kali: ~/Downloads/Labs/Facil/ENCRYPTED DATA LEAK
Session Actions Edit View Help
[v] Laboratorio desplegado correctamente.

LAB: ENCRYPTED DATA LEAK
DIFICULTAD: ★★★★★★ Fácil (1 puntos)

IP del laboratorio: 172.17.0.3

CREDENCIALES SSH:
Usuario: estudiante
Password: student2026

COMANDO DE CONEXIÓN:
ssh estudiante@localhost -p 2223
O desde otra máquina:
ssh estudiante@<IP_DEL_HOST> -p 2223

Ingresa la flag (formato FLAG{...}): FLAG{XXXXXXXXXXXXXXXXXXXX}
¡Felicitaciones! Has completado Encrypted Data Leak.

FECHA Y HORA: 2026-06-19 18:09:37 -0600

[*] Presiona Ctrl+C para eliminar el laboratorio
```